Supplementary Material: Directional Prompting for Gaze-Referential Inference in Large Multimodal Models

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This document provides the items deferred from the main paper: construction details for CG-GRI, prompt templates, and a two-part analysis of Gemini-3.1-pro's decrease in the three-candidate CG-GRI stratum.

1 Construction of CG-GRI

CG-GRI (N=483) is derived from VideoAttentionTarget [1] (N=200) and GazeFollow [2] (N=283). Neither source provides GRI-ready candidate lists, so we construct the multiple-choice options and targets as follows. We refer to the two subsets as VAT-GRI and GF-GRI; they are reporting slices of one CG-GRI track rather than separate evaluation tracks.

VideoAttentionTarget (VAT-GRI). VideoAttentionTarget consists of video clips with dense frame-level gaze-target annotations. Because consecutive frames are visually similar, we sample by gaze sequence or clip and retain at most one frame per clip. Automatic screening uses YOLOv10L [3]: the observer head box must be valid; detections overlapping the observer box are removed; the annotated gaze point must fall within at least one remaining box; and at least two candidate boxes must remain. Candidate descriptions are drafted by an LMM and human-edited so that instances of the same class remain visually distinguishable. Every retained instance is human-annotated for observer identity, candidate list, and the target indicated by the original VAT gaze point; ambiguous images are discarded. Ground-truth arrow origins are also human-annotated eye points, while endpoints remain the original VAT gaze points. This process yields 200 VAT-GRI instances.

GazeFollow (GF-GRI). GazeFollow provides web images with gaze-region annotations but no gazed-object labels. We randomly sample 1,000 images and retain those in which the observer's eyes and gaze direction can be judged. Images with occluded or unclear eyes are discarded. For every retained image, candidate objects are drafted using YOLO and human-edited, and the gazed target is human-annotated using the original GazeFollow heatmap. Ground-truth arrows use the original eye annotation as the origin and the original gaze point as the endpoint. This process yields 283 GF-GRI instances.

Shared protocol. The target observer is marked by a red unfilled bounding-box outline in every CG-GRI condition, including the RGB baseline. The box disambiguates the observer and selects whose gaze is predicted, but it is not used as the arrow origin. Among detected faces overlapping the box, we retain the prediction whose projected origin is closest to the reference head. The question lists the human-annotated candidate descriptions and asks which candidate the marked observer is looking at.

2 Prompting Templates

Plain multiple-choice prompt. This prompt is used for all image conditions unless the evaluated method explicitly changes the instructions. On CG-GRI, "the person" is replaced by "the person in the red bounding box."

```
What is the person looking at in the image?
You must choose exactly one option from the following options:
{choices}
Please answer in the following JSON format:
{"answer": "<your chosen option>",
"reasoning": "<your reasoning>"}.
```

Textual-arrow prompt.

```
What is the person looking at in the image?
The target person's gaze ray originates from pixel ({x1}, {y1}) and ends at pixel
({x2}, {y2}).
The origin (0, 0) is at the top-left corner of the image.
You must choose exactly one option from the following options:
{choices}
Please answer in the following JSON format:
{"answer": "<your chosen option>",
"reasoning": "<your reasoning>"}.
```

LMM-generated-arrow prompt. The first stage receives no answer choices and returns a gaze ray that is rendered on the image before the plain multiple-choice prompt is applied.

```
Identify the target person's gaze direction in this image.
Do not name an object.
Return exactly one JSON object using a normalized 0-1000 image coordinate system:
{"start": [x, y], "end": [x, y]}.
The start must be near the target person's visible eyes, and the end must lie in the
gaze direction.
```

The JSON `answer` field is scored by case-insensitive exact match against the ground-truth label.

3 Why Does Gemini-3.1-pro Drop in the Three-Candidate Stratum?

The main-paper stratified table shows an unusual CG-GRI result for Gemini-3.1-pro: the changes from RGB to RGB+A(UniGaze) are +2.5, −8.0, and +2.9

Table 1. Whether each model follows a wrong UniGaze arrow on three-candidate CG-GRI, among RGB→UniGaze breaks whose overlay is a wrong UniGaze arrow.

	Cases
Gemini breaks (wrong arrow, n=26)	
Followed the wrong UniGaze arrow	24
Did not follow	2
Qwen breaks (wrong arrow, n=9)	
Followed the wrong UniGaze arrow	4
Did not follow	5

percentage points for 2, 3, and 4 candidates, respectively, and only the three-candidate decrease is significant (McNemar $p<0.05$). We examine whether the newly incorrect answers in the three-candidate stratum coincide with wrong UniGaze arrows and whether the models' new answers match the distractors indicated by those arrows.

Most observed breaks coincide with a wrong UniGaze arrow. On the three-candidate CG-GRI subset, Gemini is correct with RGB but incorrect after adding A(UniGaze) for 31 instances, and Qwen for 10. We labelled each shared overlay as pointing to the ground-truth target (*correct*), a non-target candidate (*wrong*), or geometrically unclear (*ambiguous*). A wrong predicted arrow accompanies 26 of Gemini's 31 breaks (84%) and 9 of Qwen's 10 breaks (90%). The drop is therefore not unique to Gemini at the level of overlay error: both models lose RGB-correct answers mainly when the UniGaze arrow is wrong.

Gemini's observed failures align more closely with the wrong arrows. Table 1 further asks whether each model's new incorrect answer matches the distractor indicated by the wrong UniGaze arrow. Among wrong-arrow breaks, Gemini selects that distractor in 24 of 26 cases, compared with 4 of 9 for Qwen. Thus, within these observed failure cases, Gemini's errors align more closely with the erroneous directional cue. Figure 1 illustrates Gemini switching to the chair indicated by a wrong arrow while Qwen retains the RGB-correct answer; Figure 2 provides a second example in which Qwen retains the laptop despite an arrow pointing toward the dolls.

References

1. Chong, E., Wang, Y., Ruiz, N., Rehg, J.M.: Detecting attended visual targets in video. In: CVPR. pp. 5396–5406 (2020)
2. Recasens, A., Khosla, A., Vondrick, C., Torralba, A.: Where are they looking? In: NeurIPS. pp. 199–207 (2015)
3. Wang, A., Chen, H., Liu, L., Chen, K., Lin, Z., Han, J., Ding, G.: YOLOv10: Real-time end-to-end object detection. In: NeurIPS. vol. 37, pp. 107984–108011 (2024). https://doi.org/10.52202/079017-3429

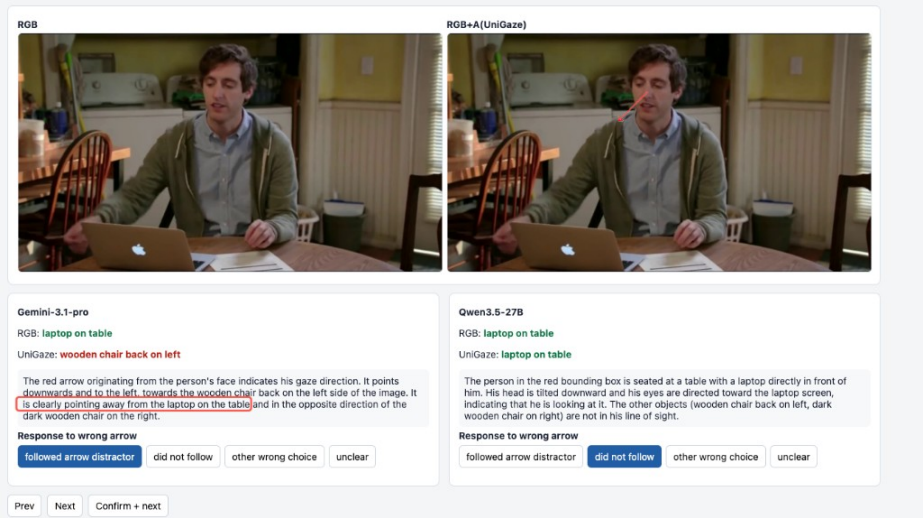


Fig. 1. Annotation view of a three-candidate instance where `Gemini-3.1-pro` follows a wrong UniGaze arrow. Ground-truth target: *laptop on table*. Both models are correct on RGB; after the overlay, Gemini switches to the left chair indicated by the arrow, while `Qwen3.5-27B` keeps the laptop.

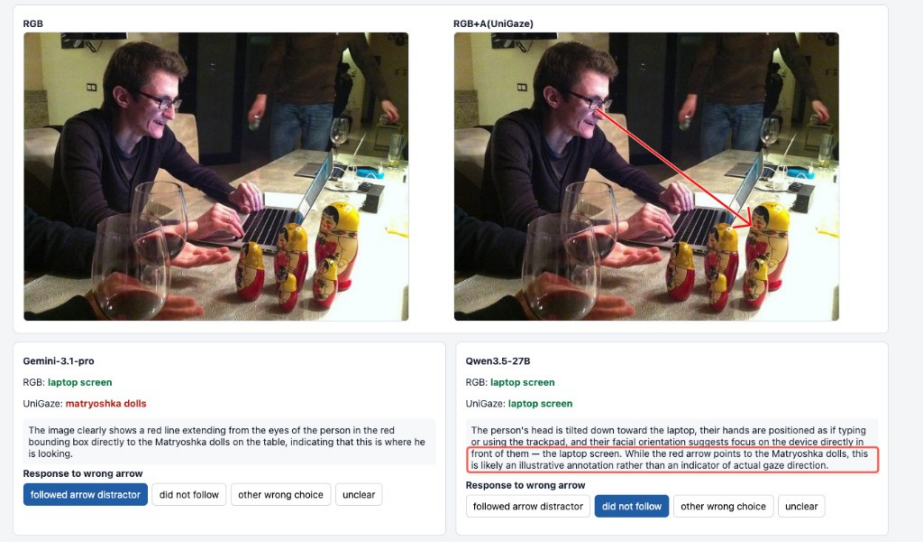


Fig. 2. Annotation view of a three-candidate instance where `Qwen3.5-27B` does not follow a wrong UniGaze arrow. Ground-truth target: *laptop screen*. The overlay points to the Matryoshka dolls. Gemini follows the arrow; Qwen keeps the laptop and treats the arrow as an illustrative annotation rather than gaze.